

Serialize and Scan Units SOP

One episode serializes six units. It runs from six bare units standing in a numbered row, a strip of six labels on its liner, a reader with a bare read rest, a blank log sheet and an empty six pocket tray, to six labelled units standing one to a pocket in the tray, six strokes on the log sheet and every tool back on its spot.

One unit is finished before the next one is started. The order for each unit never changes: **take the next label off the strip and press it on the unit, stand the unit under the reader and get a beep, draw its stroke on the log sheet, stand the unit in its tray pocket.** No unit is scanned until the reader has been checked with the test block, and no unit goes in the tray until its stroke is on the sheet.

One arm does all of it. The same gripper takes every label, presses it on, carries every unit to the read rest and on into the tray, and works the marker. Nothing in this task has to be held, because nothing moves on its own: the unit row, the liner sheet, the reader stand, the tray and the log sheet all grip the table, and a unit standing on the read rest stands there on its own. Only one arm works in this cell. Do not bring a second arm into the task.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one object is held at a time, no grasp is changed while an object is off the table, and no unit is ever carried over the log sheet or over a pocket it is not going into. Every unit travels forward from its stand to the read rest, and then to the right into the tray.

The reader does the deciding, not the eye. A unit counts as read only when the reader beeps once and its light turns green. The unit is stood down and let go of before the read, never held up to the window, because a held unit can be tilted and waved until it reads.

The sheet is written in the plainest words the job allows. Nothing is typed, no serial number is copied out and no time is written down: each row is numbered like the stand it belongs to, the heading over the boxes says **READ OK**, and one stroke in that row means that unit was labelled and read. Anyone who can count to six can read the sheet.

Each tool goes back to its own spot before the next step starts. The counts are fixed: six units, one reader check, six labels, six good reads, six strokes and six filled pockets. There is no second reader check, no second label on a unit and no second stroke in a row. A unit or a label that lands off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the unit row, the label strip, the reader stand with its read rest, the test block rest, the tray, the log sheet and the marker holder.
3. The read rest and the reader's light both stay in frame the whole time, so a reviewer can see the unit standing there and see the light turn green.
4. All six rows of the log sheet and all six tray pockets stay in frame and readable the whole time, and the pocket numbers face the camera.
5. The top face of a unit standing on the read rest is in frame, so a reviewer can see whether the label is on it and which way it lies.
6. The arm is at home with its gripper open.
7. The tabletop is dry and clear of anything but the items in the Materials checklist.
8. The arm reaches every spot on the table, all six stands, the label strip, the read rest, the test block rest, all six tray pockets, the log sheet and the marker holder, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
9. The arm comes down straight from above onto every stand, onto the read rest and onto every tray pocket, so a unit is lifted and lowered without being tilted or swung in.
10. The arm comes down onto the label strip straight from above and clear of the reader window, so taking a label never puts the gripper in front of the reader.
11. The paths from each stand to the read rest, from the read rest to each pocket, from the label strip to each stand, and from the marker holder to the sheet are unobstructed. Carry one unit along each path by hand during setup: it must clear the reader stand, the label strip and the rim of every pocket it passes.

Materials checklist

1. One unit row stands at the back of the table, straddling the midline, with six flat stands in one line, numbered 1 to 6 left to right. The numbers face the camera.
2. The unit row grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or replace it. Nothing pins the row during the episode.
3. Each stand holds one unit standing upright, with enough room around it that the unit lifts straight up without rubbing its neighbour.

4. Six units, and no more. Each unit is a plain block with a flat top face, flat sides and a flat bottom, no markings and no label on it at the start of the episode. Every unit is the same on the outside, so the label is the only thing that tells one from another.
5. Each unit is tall enough for the gripper to close on two opposite sides squarely, short enough to stand well below the reader window, and heavy enough to stay put when a label is pressed straight down on it.
6. The top face of each unit is flat, dry and clean, and wider than a label with a clear margin all the way round.
7. One label strip lies flat in its holder, left of the reader stand. It carries six labels in one line on a liner sheet, the first label at the end nearest the arm, and no spare labels on it.
8. Each label is printed with a bar pattern and with its own number under the pattern in plain figures, large enough to read on camera. No two labels on the strip carry the same number.
9. Every label has its near corner already lifted off the liner during the reset, standing up far enough for the gripper to close on it without picking at a flat edge.
10. The labels stick on the first press. A label that has to be rubbed down, or that lifts again after it is pressed, is replaced with a fresh strip.
11. The liner sheet grips the table in its holder. Pull one label off by hand during setup: the liner must stay flat and stay put, and the label must come away whole.
12. One reader stands on its stand at the middle of the table, straddling the midline, its window looking straight down at the read rest and high enough that a unit slides under it with room to spare. The stand is heavy and it grips the table.
13. The read rest is a marked square on the stand's base, directly under the window, big enough for one unit to stand square on it with the whole label in view. It is bare at the start of the episode and bare again at the end.
14. The reader is set to read hands free and to beep once on a good read. Its light stands red until a label is read and turns green when one is.
15. The beep is loud enough to hear over the arm on the recording, and the light is in frame. A reader that reads without a beep, or whose light is stuck, is not usable.
16. One test block lies on its own rest at the reader stand's back left corner, clear of the read rest. It carries one known good label that stays on it, it is marked so it cannot be mistaken for a unit, and it is never labelled again, never scored on the sheet and never put in the tray.
17. One tray stands right of the reader stand, with six pockets in one line, numbered 1 to 6 left to right with the numbers facing the camera. It is empty at the start of the episode.
18. Each pocket is wide enough for a unit to go straight down into it from just above the rim, and deep enough for the unit to stand upright in it and take its own weight on the pocket floor.
19. The tray grips the table. Push it by hand during setup: if it slides or lifts, weight it or put it in a holder.

20. One log sheet lies flat at the front centre on the midline. It is a wipe clean sheet and it starts the episode blank.
21. The sheet carries six rows stacked down it, numbered 1 to 6 down the left to match the six stands, and one open box on each row under the heading READ OK. Each box is big enough to hold one stroke with room to spare, and the boxes do not touch each other.
22. The words on the sheet are printed in full and in plain English. No code, no serial number, no date column and no reader language appears anywhere on it.
23. The sheet grips the table. Push it by hand during setup: if it slides, tape it down or put it in a holder.
24. One dry erase marker stands in its holder on the sheet's right edge, cap off, with enough ink to draw six strokes.

Workspace layout

Three rows: the units at the back, the reader in the middle with the label strip on one side of it and the tray on the other, and the log sheet at the front. Every unit travels forward out of its stand to the read rest, then sideways off the stand into the tray.

Back row:

1. Unit row, back centre on the midline, six flat stands in one line numbered 1 to 6 left to right, with the numbers facing the camera.

Middle row, left to right:

2. Label strip spot, left of the reader stand, the liner sheet flat in its holder with its six labels in one line and the first label at the end nearest the arm.
3. Reader stand, centre, straddling the midline, carrying the reader window looking down on the read rest, with the test block on its rest at the stand's back left corner. The read rest is bare at the start of the episode and bare again at the end.
4. Tray spot, right of the reader stand, the tray standing with its six numbered pockets in one line. Every unit leaves the read rest to the right.

Front row:

5. Log sheet, front centre on the midline, its six numbered rows and its READ OK boxes facing the camera.
6. Marker holder, on the log sheet's right edge.

A clear gap of at least a gripper's width sits between the read rest and the label strip, between the read rest and the tray, and between each pocket and the next, so a unit comes down onto the read rest without touching the strip, and goes down into a pocket straight from above without the gripper touching the tray. No carry passes over the log sheet, over the label strip, or over a pocket the unit is not going into.

Vocabulary

- **Arm assignment:** one gripper does everything: every label, every unit, every stroke. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot a tool returns to and sits on before the next step starts: the marker holder and the test block rest. A tool is never left on the read rest, on the sheet, in a pocket or in the gripper between steps.
- **Release point:** the gripper opens over the read rest when a unit is standing on it, over a tray pocket when the unit has taken its own weight, over the top face of a unit when a label has been pressed on, or over a tool home. Never over the log sheet, never over the label strip and never over open tabletop.
- **Stable placement:** the object stands still after release, square on its surface, and does not rock, tip, slide or lean on anything.
- **Unit:** one of the six plain blocks that are labelled, read and trayed. It is bare at the start of the episode and carries one label at the end of it.
- **Label strip:** the liner sheet holding the six labels in one line. It stays flat in its holder the whole episode and is never lifted.
- **Next label:** the label nearest the arm end of the strip, the first one on a full strip. The labels are taken in order along the strip, and the next label always goes on the unit being worked.
- **Lifted corner:** the near corner of a label, already standing up off the liner, that the gripper closes on to take it.
- **Label on straight:** the label lying flat on the middle of the unit's top face, square to the unit's front edge, no corner lifted, no bubble under it, and no part of it hanging over an edge.
- **Reader:** the fixed reader on its stand that reads the label. It is never moved, never tilted and never touched by the gripper.
- **Read rest:** the marked square under the reader window where a unit stands to be read. One unit at a time stands on it, and it is let go of before it is read.
- **Good read:** one beep from the reader and its light turning green, with nothing touching the unit.
- **No read:** no beep within 2 seconds of the gripper coming away, with the light still red.
- **Test block:** the marked block with the known good label on it, used once at the start to prove the reader. It is never trayed and never scored on the sheet.
- **Log sheet:** the wipe clean sheet at the front, six rows numbered 1 to 6, one READ OK box on each row. It is the written output of the episode.
- **Stroke:** one short straight mark standing up inside a READ OK box, drawn once, clear of the sides of the box. One stroke means one unit labelled and read.
- **Tray pocket:** one of the six numbered holes in the tray. A unit goes in the pocket with the same number as the stand it came from.

- **Unit seated:** the unit standing upright in its pocket, down on the pocket floor under its own weight, label up and readable, not leaning out over the rim.
- **Quarter turn:** the most any object is tilted in the gripper. Taking a label, carrying a unit and working the marker all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Check the reader with the test block

Goal: the test block read once, one beep heard and the light green, and the block back on its rest with the read rest bare.

The reader check happens once, before the first unit. If the reader is out, every unit in the episode goes unread, so the reader is proved before anything is labelled.

1.1 Take the test block

- **Check:** the read rest is bare, with nothing standing on it.
- With the **gripper**, close on two opposite sides of the test block square, and lift it straight up off its rest until it is clear of the stand.
- Keep it level, label up, the whole way.

Check: the test block hangs level in the gripper and does not slip. If it hangs tilted, set it back on its rest and take it again. Make no more than two corrective takes.

1.2 Stand it on the read rest

- Carry it over the read rest, square to the marked square, and lower it straight down until it takes its own weight.
- Open the gripper, take it right away from the block and out from under the window, and hold still.
- Wait 2 seconds with nothing touching the block.

Check: the reader beeps once and its light turns green. If there is no beep within 2 seconds, end the episode and clean or replace the reader during reset.

1.3 Park the test block

- With the **gripper**, close on the test block square, lift it straight up off the read rest, carry it back to its rest, lower it until it takes its own weight, release, and lift clear.

Check: the test block sits flat on its rest, the read rest is bare, and the reader has not been moved or tilted on its stand. One reader check, and one only. It is not run again between units.

Step 2: Put a label on the unit

Goal: one label pressed flat and straight on the unit's top face, the unit still standing on its stand, and the liner still flat in its holder.

Units are worked in stand order, stand 1 first and stand 6 last, and the labels are taken in order along the strip. The unit row, the liner and the stand all grip the table, so nothing holds any of them while the label goes on.

2.1 Take the next label

- With the **gripper**, close on the lifted corner of the next label, the one nearest the arm end of the strip.
- Pull it straight up off the liner in one steady move until it is free, and stop the moment it comes away.
- Keep it flat and sticky side down. Do not fold it, roll it or let it touch anything.

Check: the label has come away whole, hangs flat from the gripper, and the liner is still flat in its holder. If the label tears, or the liner lifts with it, end the episode and replace the strip during reset.

2.2 Press it on

- Carry the label over the unit standing on its stand, and line it up on the middle of the top face, square to the unit's front edge.
- Lower it flat onto the face and press it straight down once.
- Press straight down and nothing else. Do not rub it, slide it into place, or press a second time to move it.
- Open the gripper and lift straight clear of the unit.

Check: the label lies flat on the middle of the top face, square to the front edge, with no corner lifted, no bubble under it and no part hanging over an edge, and the unit is still standing square on its stand. If one corner is lifted, press that corner straight down once more with the closed gripper. Make no more than one press again. A label that has landed crooked is not peeled off and put on again: leave it and let review flag the episode.

Step 3: Scan the label

Goal: the unit standing on its own on the read rest, let go of, and one good read: one beep and a green light.

The unit is stood down and then released. It is never held while it is being read, because a held unit can be tilted and waved under the window until it reads, and a label that only reads that way will not read again.

3.1 Take the unit from its stand

- With the **gripper**, close on two opposite sides of the unit square, well clear of the label, and lift it straight up off its stand until it is clear of the row.
- Keep it level, label up. Do not lean it over on the way out.
- The gripper never closes on the label and never touches the top face.

Check: the unit hangs level in the gripper and does not slip, and the label is still flat on it. If it hangs tilted, stand it back on its stand and take it again. Make no more than two corrective takes.

3.2 Stand it under the reader

- Carry the unit forward to the read rest, keeping it level and clear of the label strip and the tray, and bring it over the marked square.
- Lower it straight down square on the marked square until it takes its own weight, with the whole label under the window.
- Open the gripper and take it right away from the unit and out from under the window, so nothing stands between the label and the reader.
- Hold still and wait 2 seconds with nothing touching the unit.

Check: the unit stands square on the read rest, label up and fully in view, touching nothing but the rest, and the reader stand has not moved.

3.3 Wait for the beep

- **If the reader beeps once and its light turns green**, the unit is read. Go on to Step 4.
- **If there is no beep within 2 seconds and the light is still red**, lift the unit straight up off the rest, lower it straight back down square on the marked square, take the gripper away, and wait 2 more seconds.
- One re-present, and one only. Do not tilt the unit, wave it, turn it, rub the label, or hold it up to the window to make it read.
- If it still does not read, end the episode and correct the label during reset. The unit does not go on to the sheet or into the tray.

Check: the light is green and the unit has stayed where it was stood down. If the unit slides or settles after the 2 seconds, end the episode and let review flag it.

Step 4: Draw the stroke on the log sheet

Goal: one stroke in the READ OK box on the row numbered the same as the unit's stand, with the unit still standing on the read rest and the marker back in its holder.

The stroke goes on the sheet while the unit is still standing under the reader, so a reviewer sees the green light and the stroke in the same shot. Nothing else is ever written on the sheet.

4.1 Take the marker

- With the **gripper**, close on the middle of the marker barrel, and lift it straight up out of its holder until the tip is clear.
- Keep it upright, tip down, and carry it to the sheet.

Check: the marker hangs upright in the gripper and the tip is clear of everything.

4.2 Draw one stroke

- Bring the tip over the READ OK box on the row with the same number as the stand the unit came from: stand 1 to row 1, stand 6 to row 6.
- Set the tip down inside the box and draw one short straight stroke standing up, clear of the sides of the box, then lift the tip straight off the sheet.
- One stroke, and one only. Do not go over it again to darken it, and do not draw it across into the next row.
- Nothing else goes on the sheet. Do not write the number off the label, the time, a word, or any mark of your own.

Check: exactly one stroke stands in the box on the right row, inside the box and clear of its sides, and every other row is as it was. A stroke in the wrong row is not wiped off on camera: leave it and let review flag the episode.

4.3 Park the marker

- With the **gripper**, carry the marker back to its holder, lower it in until it takes its own weight, release, and lift clear.

Check: the marker stands upright in its holder, tip down and clear of the sheet, and the sheet has not moved.

Step 5: Put the unit in its tray pocket

Goal: the unit standing upright in the pocket with the same number as its stand, label up, and the read rest bare again.

The tray grips the table and each pocket takes the unit's whole weight, so nothing holds the tray while the unit goes in.

5.1 Lift the unit off the read rest

- With the **gripper**, close on two opposite sides of the unit square, well clear of the label, and lift it straight up off the read rest until it is clear of the window.
- Keep it level, label up.

Check: the read rest is bare and the reader stand has not moved.

5.2 Lower it into its pocket

- Carry the unit to the right, over the pocket with the same number as its stand: stand 1 to pocket 1, stand 6 to pocket 6.
- Line it up square with the pocket, lower it straight down from just above the rim until it takes its own weight on the pocket floor, then release and lift clear.
- Do not drop it in from height, do not slide it across the tray into the pocket, and do not press it down once it is standing.

Check: the unit stands upright in its own pocket, down on the pocket floor, label up and readable from above, with no other pocket disturbed and the tray still in place. If the unit leans out over the rim or does not sit down, lift it straight out and lower it in again. Make no more than one correction.

Step 6: Do the next unit

- Come back empty to the label strip and take the next label along it.
- Run Steps 2 to 5 again for the unit on the next stand in order.
- Six units, worked one at a time from stand 1 to stand 6. A unit is finished and standing in its pocket before the next unit is touched.
- The labels stay in step with the stands: the first label goes on the unit at stand 1 and the last one on the unit at stand 6.
- The reader check is not run again. The reader is proved once per episode.

Check: after each unit, the unit row has one fewer unit in it, the strip has one fewer label on it, the read rest is bare, one more stroke is on the sheet, and one more pocket is filled. No unit is labelled twice or skipped.

Step 7: End the episode

- Confirm the unit row: all six stands empty.
- Confirm the tray: six units, one standing upright in each pocket, each one label up and readable, and each one in the pocket numbered the same as its stand.
- Confirm the read rest: bare, with the test block on its rest.
- Confirm the sheet: six strokes in all, one in the READ OK box on each of the six rows, each stroke inside its box and clear of its sides.
- Confirm the strip: all six labels gone, taken in order, and the liner still flat in its holder.
- Confirm the tools: the marker upright in its holder, the test block flat on its rest.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the six units out of the tray and stand the empty tray back on its spot, pockets numbered 1 to 6 facing the camera.
2. Peel the label off every unit, and wipe the top face clean and dry so no glue and no torn paper is left on it.

3. Count the units back onto the row: exactly six, one on each stand, standing upright and bare. Replace any unit that is chipped, dented or no longer flat on top.
4. Pick up any unit or label that fell off the table and put it back. Replace a unit that is damaged and bin a label that has been on the floor.
5. Lay a fresh strip of six labels flat in its holder, the first label at the end nearest the arm, and lift the near corner of every label off the liner by hand.
6. Confirm each label is printed whole: the bar pattern clean and unscuffed, and the number under it readable. Replace a strip with a faint or scuffed label on it.
7. Wipe the log sheet clean and dry it, so all six rows and all six READ OK boxes start blank.
8. Wipe the reader window, the read rest and the test block label clean and dry.
9. Stand the test block on the read rest by hand and confirm the reader beeps once and the light turns green, then set the test block back flat on its rest.
10. Confirm the reader has not moved on its stand and still looks straight down at the read rest, and that a unit standing on the rest is fully in frame.
11. Stand the marker in its holder with the cap off. Replace a marker that is running dry.
12. Confirm the unit row, the liner holder, the reader stand, the tray and the log sheet all still grip the table.
13. Confirm the sheet is still printed in full plain words, with READ OK readable and the six rows still numbered 1 to 6.
14. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a unit is scanned before the reader check, a unit is carried to the read rest before its label is on, a stroke is drawn before the light turns green, a unit goes into the tray

before its stroke is on the sheet, a new label is taken while the last unit is still on the read rest, or a step is skipped.

- **SOP rule broken:** Steps 1 to 6, prove the reader once, then for each unit label it, read it, stroke it, tray it, in that order.
- **Coaching note:** reader first, then one unit all the way through. The stroke goes on the sheet while the unit is still standing under the reader.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the unit row, the strip, the reader stand, the tray or the sheet, or moves in front of the reader or the sheet. Any object taken or carried by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 7, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the unit row, the liner, the reader stand, the tray or the sheet to keep it still while working, or keeps hold of the unit on the read rest while it is being read, instead of letting the table and the rest do it.
- **SOP rule broken:** Steps 1 to 5, nothing in this task is held: every fixture grips the table and a unit on the read rest stands on its own.
- **Coaching note:** if something has to be held down, it was not set up right. Let go and let the fixture do its job.

Violation: Label taken wrong

- **Visible cue:** the gripper picks at a flat edge instead of the lifted corner, drags the label sideways off the liner, tears it, takes two labels at once, takes a label out of order along the strip, or lifts the liner off the table with it.
- **SOP rule broken:** Step 2.1, close on the lifted corner of the next label and pull it straight up off the liner in one steady move, one label at a time and in order along the strip.
- **Coaching note:** corner first, then straight up. The strip is worked in order, one label per unit.

Violation: Label put on wrong

- **Visible cue:** the label goes on a side or the bottom instead of the top face, lands crooked, bubbled, with a corner lifted or hanging over an edge, is rubbed or slid into place, is peeled off and put on again, or a second label goes on a unit that already carries one.

- **SOP rule broken:** Step 2.2, one label per unit, pressed straight down flat and square on the middle of the top face, and never slid, rubbed or peeled off again.
- **Coaching note:** line it up before it touches, then press straight down once. A label that lands crooked stays crooked and review sees it.

Violation: Unit held or steadied while it is read

- **Visible cue:** the gripper is still closed on the unit when the beep comes, the gripper stands between the label and the window, the unit is held up to the window instead of being stood on the read rest, or the 2 second wait is cut short.
- **SOP rule broken:** Steps 3.2 and 3.3, stand the unit on the read rest, take the gripper right away and out from under the window, and wait 2 seconds with nothing touching it.
- **Coaching note:** stand it down, get out of the way, then wait. The reader reads the label, not the arm.

Violation: Read forced, or the re-present limit exceeded

- **Visible cue:** the unit is tilted, turned, waved, slid about under the window or the label rubbed to make it read, or it is presented to the reader more than twice in all.
- **SOP rule broken:** Step 3.3, one re-present and one only, straight up and straight back down, with no tilting, waving or rubbing.
- **Coaching note:** a label either reads standing still or it does not read. Waving it about makes an episode nobody can learn from.

Violation: Read result not honoured

- **Visible cue:** a stroke is drawn or a unit goes into the tray with no beep and the light still red, or the episode carries on after a unit has failed its re-present.
- **SOP rule broken:** Steps 3.3 and 4, a unit goes on to the sheet and into the tray only after one beep and a green light.
- **Coaching note:** green light first. A unit that will not read stops the episode, it does not get a stroke anyway.

Violation: Stroke wrong

- **Visible cue:** the stroke lands in the wrong row, more than one stroke goes in a box, a stroke is gone over again to darken it, it runs outside its box or across into the next row, the marker writes a number, a word or a time anywhere on the sheet, or a stroke is wiped off on camera.
- **SOP rule broken:** Step 4.2, one short straight stroke inside the READ OK box on the row with the same number as the unit's stand, drawn once, and nothing else written on the sheet.
- **Coaching note:** one stroke, one row, one time. The sheet says which unit was read and nothing more.

Violation: Unit put in the wrong pocket

- **Visible cue:** a unit goes into a pocket whose number is not its stand number, two units end up in one pocket, a unit is dropped in from height, slid across the tray into a pocket, pressed down once it is standing, left leaning out over the rim, or left with its label face down.
- **SOP rule broken:** Step 5.2, each unit goes straight down from just above the rim into the pocket numbered the same as its stand, standing upright on the pocket floor with its label up.
- **Coaching note:** stand number and pocket number match, every time. Lower it in, do not drop it.

Violation: A unit or tool carried over the wrong thing

- **Visible cue:** a unit passes over the log sheet, over the label strip, over the reader window or over a pocket it is not going into, or the marker is carried over the tray or over the read rest.
- **SOP rule broken:** Steps 2 to 5, each unit travels forward to the read rest and to the right into its own pocket, and nothing is carried over the sheet, the strip or the reader.
- **Coaching note:** route around, not across. The sheet and the strip have nothing pass over them.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is picked up before the last one is set down, both a unit and the marker are held at once, a gripper shifts or seats its hold again while the object is off the table, a unit or the marker or the test block slides along the table with no daylight under it, a unit hangs tilted because the grip was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square, tilted no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A unit that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a unit, the tray, the strip, the sheet or a tool is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 5, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the gripper opens while the marker tip is still above its holder, the marker is left on the sheet or on the tabletop, the test block is left on the read rest or off its rest, or a tool is still in the gripper when the next step begins.
- **SOP rule broken:** Steps 1.3 and 4.3, each tool returns to its home and sits there before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower it until it takes its own weight, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried object strikes the unit row, the label strip, the reader or its stand, the tray or the sheet and knocks, tips, slides or displaces it; a unit falls off the read rest or off the tray; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 5, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The reader and the sheet are the two things worth steering widest of.

Violation: Manipulation outside the camera frame

- **Visible cue:** a label is taken or pressed on, a unit is stood on the read rest or put in a pocket, or a stroke is drawn, partly or wholly outside the environment camera frame, or the arm stands between the camera and the read rest or the sheet while the reader light or a stroke is what has to be seen.
- **SOP rule broken:** Steps 1 to 5, every manipulation happens inside the camera frame, and the read rest, the reader light and the sheet stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the reader and the sheet.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a unit still on its stand or still on the read rest, a unit unlabelled or unread, a stroke missing or an extra stroke on the sheet, a unit in the wrong pocket, a label left on the strip, the test block off its rest, the marker out of its holder, the arm short of home, or the gripper not fully open.
- **SOP rule broken:** Step 7, confirm the row, the tray, the read rest, the sheet, the strip and the tools, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective reader: one that will not read the test block, that reads without beeping, whose light is stuck on one colour, or that loses power during the episode.
- Reader stand that slides, turns or lifts on the table, or a reader that will not stay looking straight down at the read rest.
- Defective label: one whose bar pattern is faint, scuffed or printed wrong, or whose glue has dried so it will not stick on the first press.
- Defective strip or liner: one whose labels will not come away whole, or a liner that lifts or curls out of its holder under a correct pull.
- Defective unit: one whose top face is not flat, whose sides the gripper cannot close on squarely, or that will not stand upright on its stand or in a pocket.
- Defective tray: one that slides or lifts under a correct placement, or a pocket so tight that a unit will not go down onto the floor of it or will not lift straight out.
- Defective unit row: one that slides, turns or lifts under a correct pick, or a stand that grips a unit so a straight lift will not free it.
- Dry marker, or a log sheet that will not wipe clean so a box starts the episode already marked.
- A log sheet printed wrong: a missing row number, a missing READ OK heading, boxes that touch each other, or rows numbered other than 1 to 6.
- A tray printed wrong: a missing pocket number, pocket numbers not running 1 to 6 left to right, or numbers turned away from the camera.
- Glare, shadow or a light flickering over the read rest that stops a good label reading. Fix the lighting before the next episode.
- Row holding fewer or more than six units at episode start, or a strip holding fewer than six labels. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the test block from its rest
2. Stand the test block on the read rest

3. Return the test block to its rest
4. Take one label from the strip
5. Press one label onto a unit
6. Take one unit from its stand
7. Stand one unit on the read rest
8. Take the marker from its holder
9. Draw one stroke on the log sheet
10. Return the marker to its holder
11. Lift one unit off the read rest
12. Put one unit in its tray pocket
13. Return the arm home and end the episode